CRUD

```
Package com.project.utility;

Import java.sql.Connection;
Import java.sql.DriverManager;
Import java.sql.PreparedStatement;
Import java.sql.ResultSet;
Import java.sql.Statement;
Import java.util.Scanner;

Public class DbConnection {

	Public static void main(String[] args) {
//		create();
//		insert();
//		update();
//		delete();
		Search();
//		showAll();
	}

	Private static void create() {

		Connection con = null;
		PreparedStatement ps = null;
		Try {
			Class.forName("com.mysql.cj.jdbc.Driver");
			System.out.println("Driver loaded…");
			Con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/d5578","root","j0j
0p@ssw0rd");
			If(con!=null)
				System.out.println("connection done");
			Else
				System.out.println("Not connected..");
```

```
                        Ps = con.prepareStatement("create table tbl_city(city_id
numeric(10) primary key, cname varchar(30)) ");
                        Ps.executeUpdate();
                        System.out.println("Table created");

//                      Statement st = con.createStatement();
//                      st.executeUpdate("create table tbl_city(city_id numeric(10)
primary key, cname varchar(30))");
//                      System.out.println("Table created..");

                }catch(Exception e) {
                        e.printStackTrace();
                }finally {
                        Try {
                                Ps.close();
                                Con.close();
                                System.out.println("Connections closed from the
finally block");
                        }catch(Exception e) {
                                e.printStackTrace();
                        }
                }

        }

        Private static void insert() {
                Connection con = null;
                PreparedStatement ps = null;
                Scanner sc = null;
                Try {
                        Sc = new Scanner(System.in);
                        Class.forName("com.mysql.cj.jdbc.Driver");
                        System.out.println("Driver loaded…");
                        Con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/d5578","root","j0j
0p@ssw0rd");
                        If(con!=null)
```

```
                System.out.println("connection done");
            Else
                System.out.println("Not connected..");

            Ps = con.prepareStatement("insert into tbl_city values(?,?)
");
            System.out.println("Enter city id");
            Int id = sc.nextInt();
            Ps.setInt(1, id);
            System.out.println("Enter city name");
            String city = sc.next();
            Ps.setString(2, city);

            Int rows = ps.executeUpdate();
            If(rows>0) {
                System.out.println("Record inserted successfully");
            }else {
                System.out.println("Something went wrong to insert
record");
            }

        }catch(Exception e) {
            e.printStackTrace();
        }finally {
            Try {
                If(ps!=null)
                    Ps.close();
                If(con!=null)
                    Con.close();
                If(sc!=null)
                    Sc.close();
                System.out.println("Connections closed from the
finally block");
            }catch(Exception e) {
                e.printStackTrace();
            }
        }
```

```
        }

        Private static void update() {
                Connection con = null;
                PreparedStatement ps = null;
                Scanner sc = null;
                Try {
                        Sc = new Scanner(System.in);
                        Class.forName("com.mysql.cj.jdbc.Driver");
                        System.out.println("Driver loaded…");
                        Con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/d5578","root","j0j
0p@ssw0rd");
                        If(con!=null)
                                System.out.println("connection done");
                        Else
                                System.out.println("Not connected..");

                        Ps = con.prepareStatement("update tbl_city set cname=?
Where city_id=? ");
                        System.out.println("Enter city id to update");
                        Int id = sc.nextInt();
                        Ps.setInt(2, id);
                        System.out.println("Enter city name");
                        String city = sc.next();
                        Ps.setString(1, city);

                        Int rows = ps.executeUpdate();
                        If(rows>0) {
                                System.out.println("Record updated successfully");
                        }else {
                                System.out.println("Something went wrong to update
record");
                        }

                }catch(Exception e) {
                        e.printStackTrace();
```

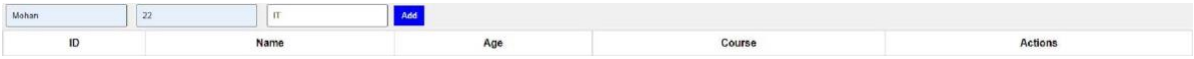
```
		}finally {
			Try {
				If(ps!=null)
					Ps.close();
				If(con!=null)
					Con.close();
				If(sc!=null)
					Sc.close();
				System.out.println("Connections closed from the
finally block");
			}catch(Exception e) {
				e.printStackTrace();
			}
		}
	}

	Private static void delete() {
		System.out.println("delete");
	}

	Private static void search() {
		Connection con = null;
		PreparedStatement ps = null;
		Scanner sc = null;
		Try {
			Class.forName("com.mysql.cj.jdbc.Driver");
			System.out.println("Driver loaded…");
			Con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/d5578","root","j0j
0p@ssw0rd");
			If(con!=null)
				System.out.println("connection done");
			Else
				System.out.println("Not connected..");

			Ps = con.prepareStatement("select * from tbl_city where
city_id = ?");
```

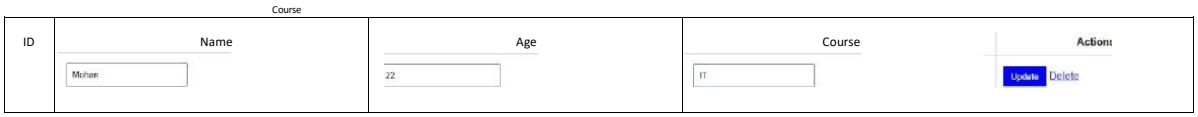
```
                Sc = new Scanner(System.in);
                System.out.println("Enter city id to search");
                Int id = sc.nextInt();
                Ps.setInt(1, id);

                ResultSet rs = ps.executeQuery();
                While(rs.next()) {
                        Int city_id = rs.getInt(1);
                        String city = rs.getString(2);
                        System.out.println("City id: "+city_id+" City Name:
"+city);
                }


        }catch(Exception e) {
                e.printStackTrace();
        }finally {
                Try {
                        If(ps!=null)
                                Ps.close();
                        If(con!=null)
                                Con.close();
                        System.out.println("Connections closed from the
finally block");
                }catch(Exception e) {
                        e.printStackTrace();
                }
        }
    }

    Private static void showAll() {
        Connection con = null;
        PreparedStatement ps = null;
        Try {
                Class.forName("com.mysql.cj.jdbc.Driver");
                System.out.println("Driver loaded…");
```

```
                        Con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/d5578","root","j0j
0p@ssw0rd");
                        If(con!=null)
                                System.out.println("connection done");
                        Else
                                System.out.println("Not connected..");

                        Ps = con.prepareStatement("select * from tbl_city ");

                        ResultSet rs = ps.executeQuery();
                        While(rs.next()) {
                                Int city_id = rs.getInt(1);
                                String city = rs.getString(2);
                                System.out.println("City id: "+city_id+" City Name:
"+city);
                        }


                }catch(Exception e) {
                        e.printStackTrace();
                }finally {
                        Try {
                                If(ps!=null)
                                        Ps.close();
                                If(con!=null)
                                        Con.close();
                                System.out.println("Connections closed from the
finally block");
                        }catch(Exception e) {
                                e.printStackTrace();
                        }
                }
        }

}
```

Create:

Student Manager (CRUD)

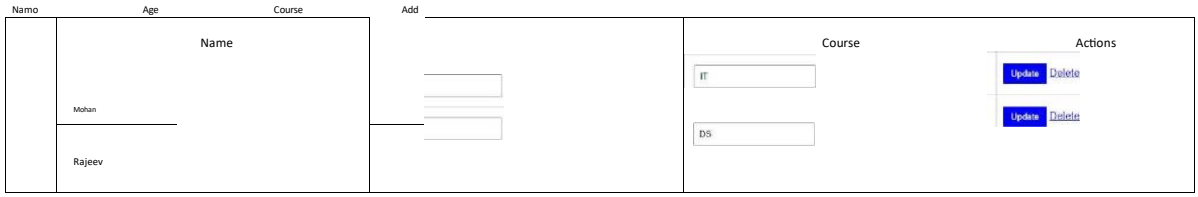




Update:

Student Manager (CRUD)

Student Manager (CRUD)



Delete:

Student Manager (CRUD)

Student Manager (CRUD)

```
$ sqlite3 students.db
SQLite version 3.45.0 2024-01-15
Enter ".help" for usage hints.

sqlite> CREATE TABLE IF NOT EXISTS students (
   ...> id INTEGER PRIMARY KEY AUTOINCREMENT,
   ...> name TEXT NOT NULL,
   ...> age INTEGER,
   ...> course TEXT
   ...> );
sqlite>

sqlite> .schema students
CREATE TABLE students (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
```

name TEXT NOT WLL.

age INTEGER,

. . . > course TEXT

sqlitey

```
sqlite> - schema students
CREATE TABLE students id INTEGER PRIMARY KEY
    AUTOINCRE'OENT, nane TEXT NOT NULL, age
    INTEGER, course TEXT

sqlite> INSERT INTO students (name,    age,  course) VALUES ( 'Rahul • ,    ' Convuter Science ' )
                                              21,                           ;
sqlite> INSERT INTO students (name,    age,  course) VALUES ( 'Anita' 22,
sqlite> INSERT INTO Students (name,    age,  course) VALUES ('John's 2Ø,
sq1 ite>

sqlite>  SELECT  *  FR(Y'I  students;
llRahul    1211Computer    Science
21Anita12211T | 29 | Al
sqlite> UPDATE students SET naræ—— Rahul Sharma' age—22, course— 'Data Science• I,HERE id-l; sqlite>

sqlite»   SELECT   FRW   students;   11Rahu1
Sharma 1221Data Science
21Anita12211T
3 pohn

sqlite»  DELETE  FROM  students  b.å4ERE  id=2;
sq1 ite>

sqlite>  SELECT  *  FRa•1  students;  1
IRahu1   Sharma   1221Data   Science
3Pohn1281A1


sqlite> INSERT INTO students (name. age, course) VALUES ( •Priya , 23 , • Cyber Security'); sqlite>

sqlite>  SELECT  *  FRCP)  students;  1
IRahu1 Sharma 1221 Data Science
3 loohnl
41priya123 Icyber Security

sqlite> -exit
```